Q1. A company needs is granted the site address 201.70.64.0. The company needs six subnets. Design the subnets using cisco packet tracer and test the connectivity using ping command. (Intermediate device – Router)

A. Subnetting Calculation

1. Subnet Information

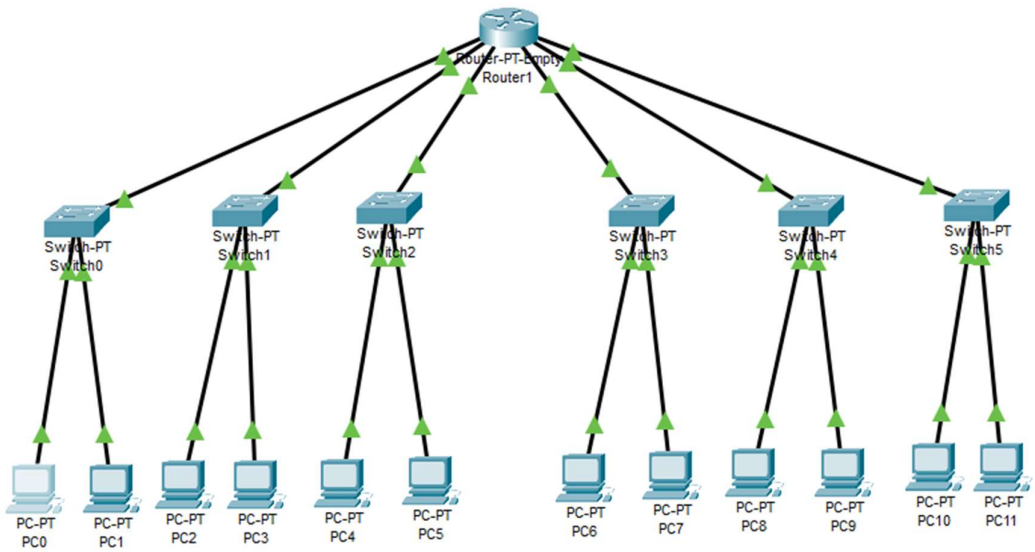
- **Given address:** 201.70.64.0 (Class C)
- **Needed subnets:** 6 (choose 8 for power of 2)
- **New subnet mask:**
  - Borrow 3 bits: $2^3 = 8$ subnets
  - Mask: 255.255.255.224 (/27)

2. Subnet Ranges Table

Subnet	Network Address	Usable IP Range	Broadcast
1	201.70.64.0	201.70.64.1 – 201.70.64.30	201.70.64.31
2	201.70.64.32	201.70.64.33 – 201.70.64.62	201.70.64.63
3	201.70.64.64	201.70.64.65 – 201.70.64.94	201.70.64.95
4	201.70.64.96	201.70.64.97 – 201.70.64.126	201.70.64.127
5	201.70.64.128	201.70.64.129 – 201.70.64.158	201.70.64.159
6	201.70.64.160	201.70.64.161 – 201.70.64.190	201.70.64.191

B. Topology Design

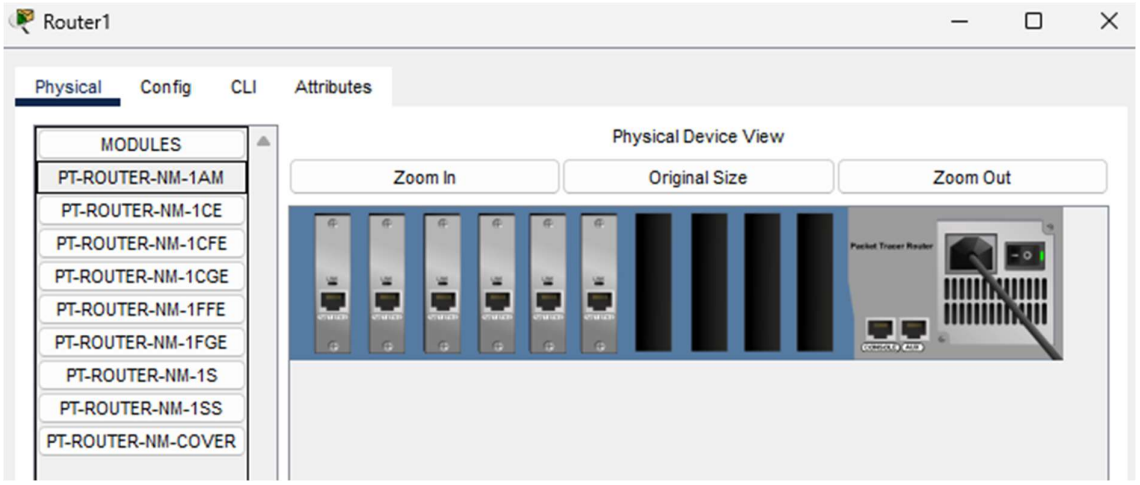
- **Place 1 router (with multiple LAN interfaces, e.g., 2811)**
- **Place 6 switches**
- **Place at least 12 PCs (2 for each subnet)**
- **Connect:**
  - Each router FastEthernet to one switch
  - 2 PCs to each switch



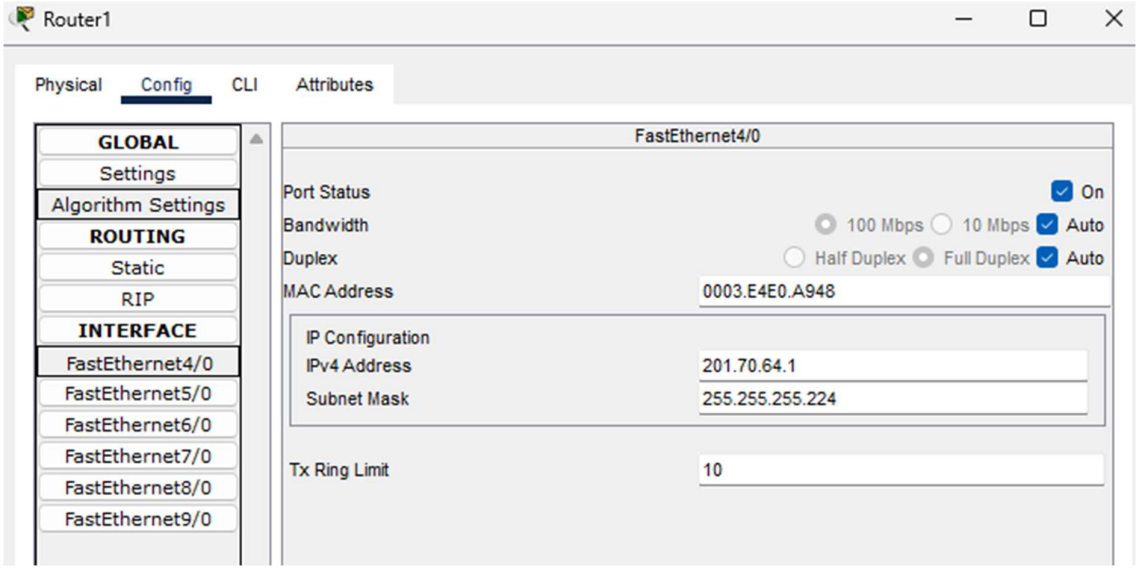
C.

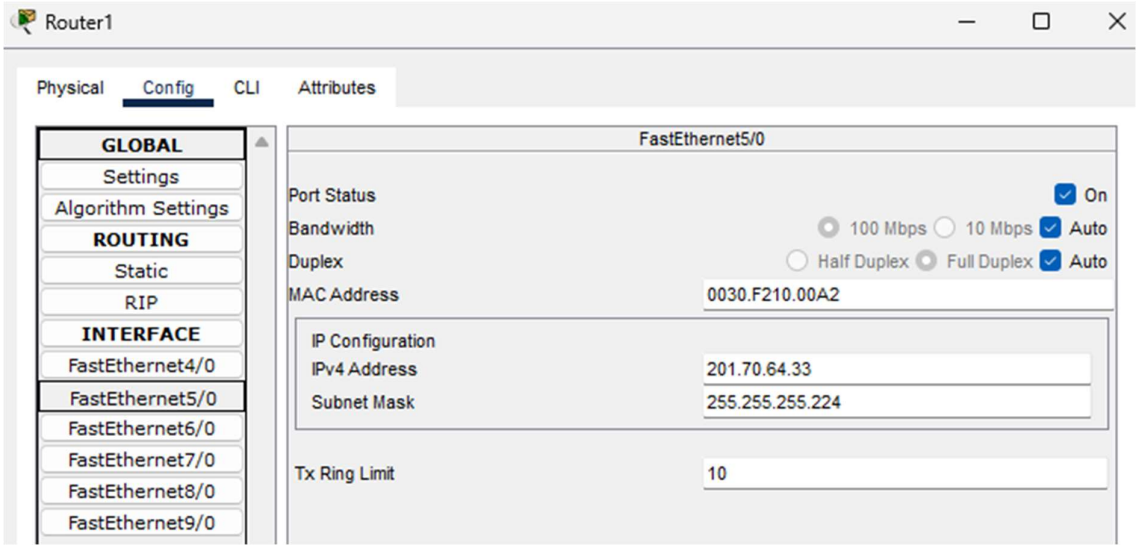
Configure Router LAN Interfaces (GUI Steps for each subnet)

For each subnet, repeat:



1. Click the **Router → Config tab**.
2. Click interface (e.g., FastEthernet0/0, FastEthernet0/1, etc.).
3. Assign the **IP Address** for each subnet:
    - Fa4/0: 201.70.64.1, mask 255.255.255.224
    - Fa5/0: 201.70.64.33, mask 255.255.255.224
    - Fa6/0: 201.70.64.65, mask 255.255.255.224
    - ...Continue for all 6 LAN interfaces.
4. Turn **Port Status** ON.



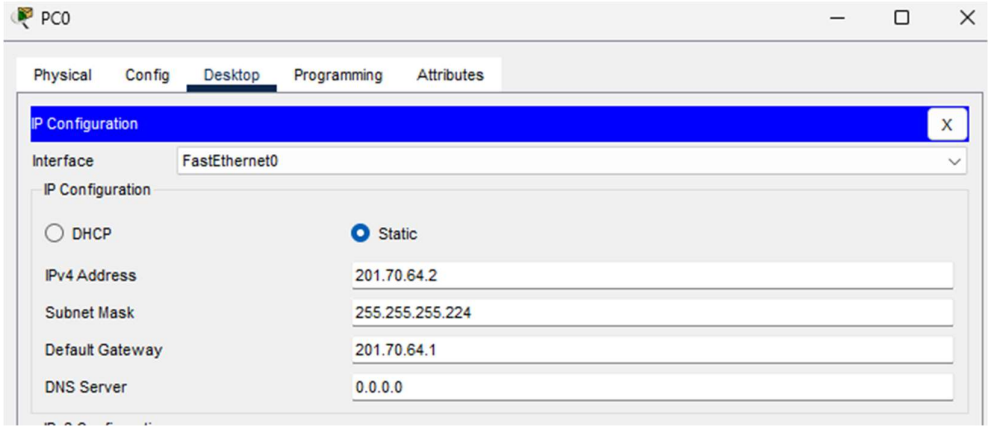


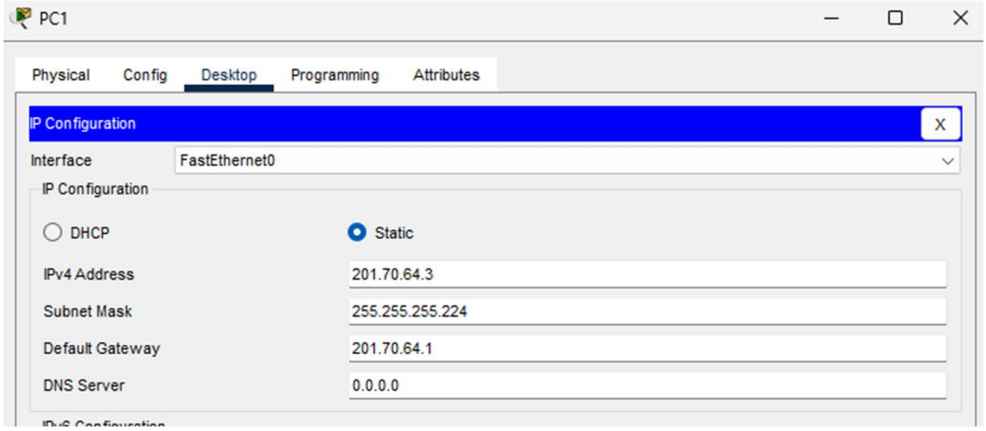
(If your router has fewer than 6 FE interfaces, use sub-interfaces with VLANs or add router modules as needed.)

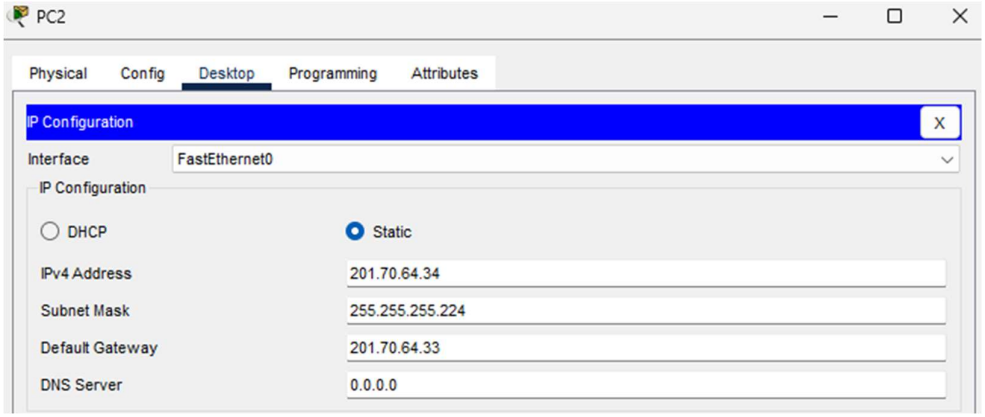
D. Configure PCs with IP Addresses (GUI Steps)

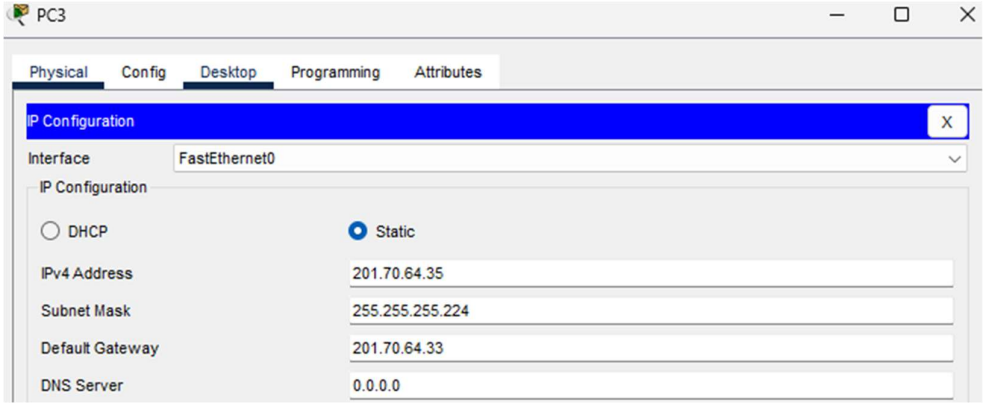
For each PC connected to each switch:

1. Click the **PC** → Desktop tab → **IP Configuration**.
2. Assign:
    - **IP Address:** Use a usable IP in the subnet (e.g., 201.70.64.2 for first subnet)
    - **Subnet Mask:** 255.255.255.224
    - **Default Gateway:** The router's corresponding interface IP (e.g., 201.70.64.1 for first subnet)


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
201.70.64.2
Subnet Mask
255.255.255.224
Default Gateway
201.70.64.1
DNS Server
0.0.0.0

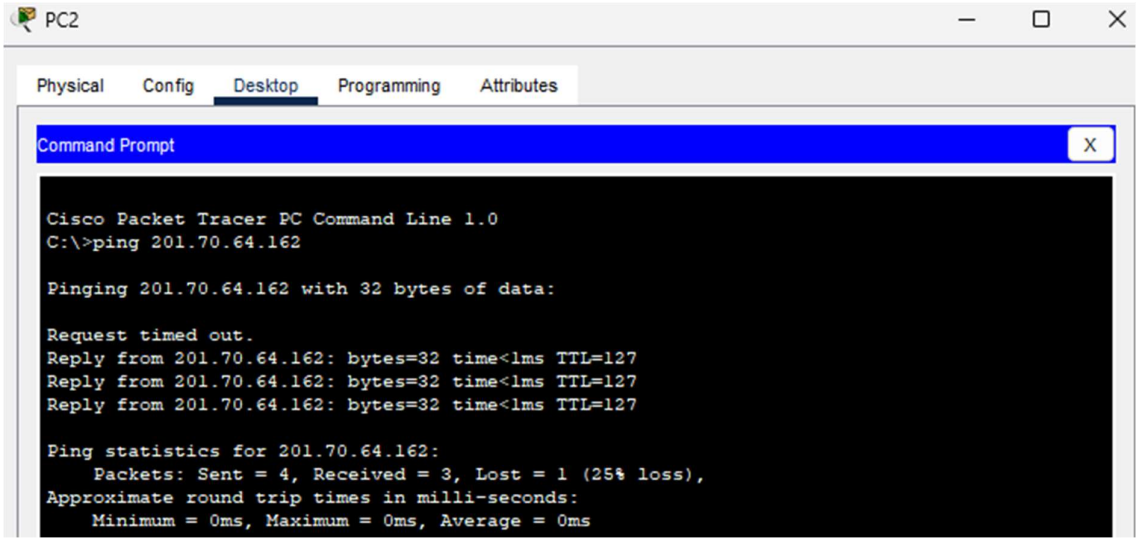

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
201.70.64.3
Subnet Mask
255.255.255.224
Default Gateway
201.70.64.1
DNS Server
0.0.0.0


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
201.70.64.34
Subnet Mask
255.255.255.224
Default Gateway
201.70.64.33
DNS Server
0.0.0.0


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
201.70.64.35
Subnet Mask
255.255.255.224
Default Gateway
201.70.64.33
DNS Server
0.0.0.0

E. Testing the Configuration

1. On a PC in each subnet, open **Command Prompt** (Desktop tab).
2. Ping:
    - The **default gateway** (router's interface).
    - Another PC in a different subnet (e.g., from 201.70.64.2 to 201.70.64.34).



Summary Table of Router GUI Static IP Assignments

Interface	IP Address	Subnet Mask	Connected Switch/Subnet
FastEthernet4/0	201.70.64.1	255.255.255.224	Switch 1 / Subnet 1
FastEthernet5/0	201.70.64.33	255.255.255.224	Switch 2 / Subnet 2
FastEthernet6/0	201.70.64.65	255.255.255.224	Switch 3 / Subnet 3
FastEthernet7/0	201.70.64.97	255.255.255.224	Switch 4 / Subnet 4
FastEthernet8/0	201.70.64.129	255.255.255.224	Switch 5 / Subnet 5
FastEthernet9/0	201.70.64.161	255.255.255.224	Switch 6 / Subnet 6

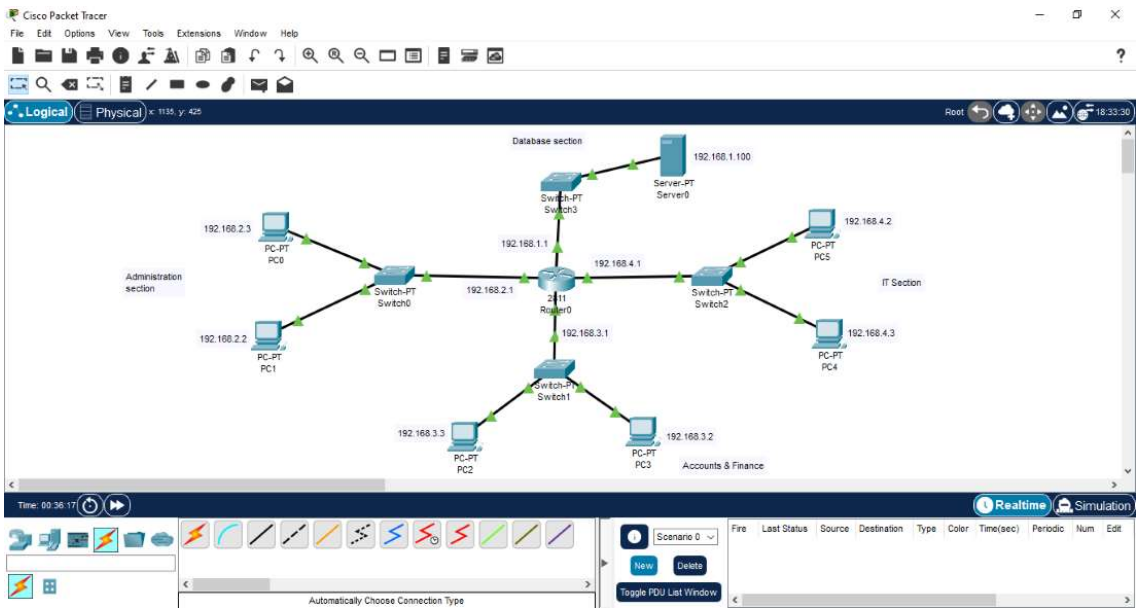
Q2: Small Organizational Setup in CISCO Packet Tracer

Step 1: Open the Cisco Packet Tracer.
Step 2.: After opening the Cisco Packet tracer, add a router, 4 switches, 6 PCs, and a server to build a network for a small organization.

Assume there're four sections in this small organization:

1. Administration section
2. Accounts and Finance
3. Information Technology (IT).
4. Database section.
There are four different networks in this organization:

1. 192.168.1.0/24
2. 192.168.2.0/24
3. 192.168.3.0/24
4. 192.168.4.0/24.

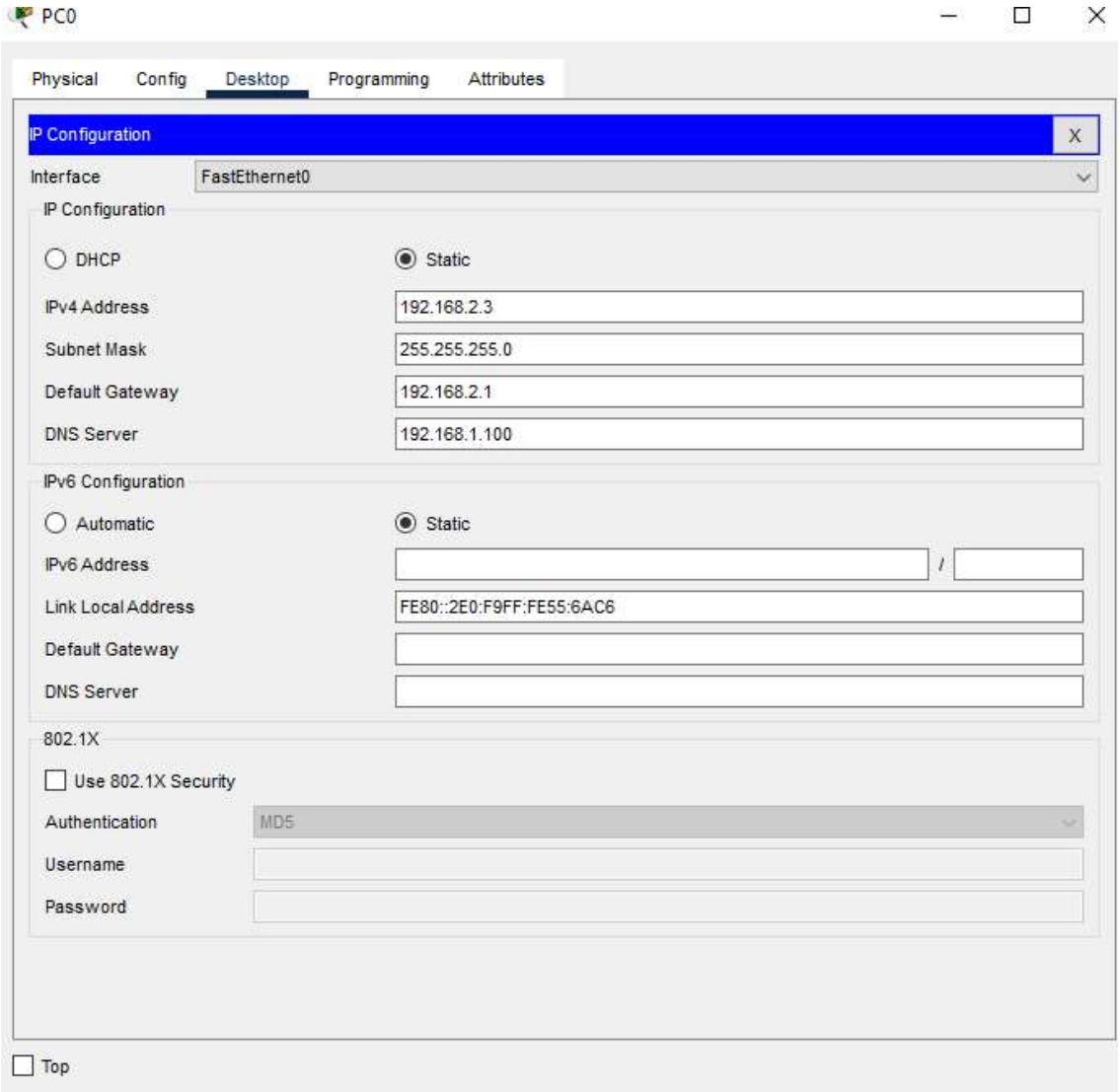


Cisco packet tracer

Step 3: Connect the router with 4 switches, 3 switches are connected with 2PCs each, and 1 switch is connected with the server using a cable. There are four different networks in this organization:- 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24, and 192.168.4.0/24.
Step 4: Give IP, subnet mask, default gateway, and DNS server to each PC and server in this network. To assign IP to each PC and server, click on each PC, go to Desktop, and then click on IP configuration.

Components	IP Address	Subnet Mask	Default Gateway	DNS server
PC0	192.168.2.3	255.255.255.0	192.168.2.1	192.168..1.100
PC1	192.168.2.2	255.255.255.0	192.168.2.1	192.168.1.100
PC2	192.168.3.3	255.255.255.0	192.168.3.1	192.168.1.100
PC3	192.168.3.2	255.255.255.0	192.168.3.1	192.168.1.100
PC4	192.168.4.3	255.255.255.0	192.168.4.1	192.168.1.100
PC5	192.168.4.2	255.255.255.0	192.168.4.1	192.168.1.100
Server	192.168.1.100	255.255.255.0	192.168.1.1	192.168.1.100

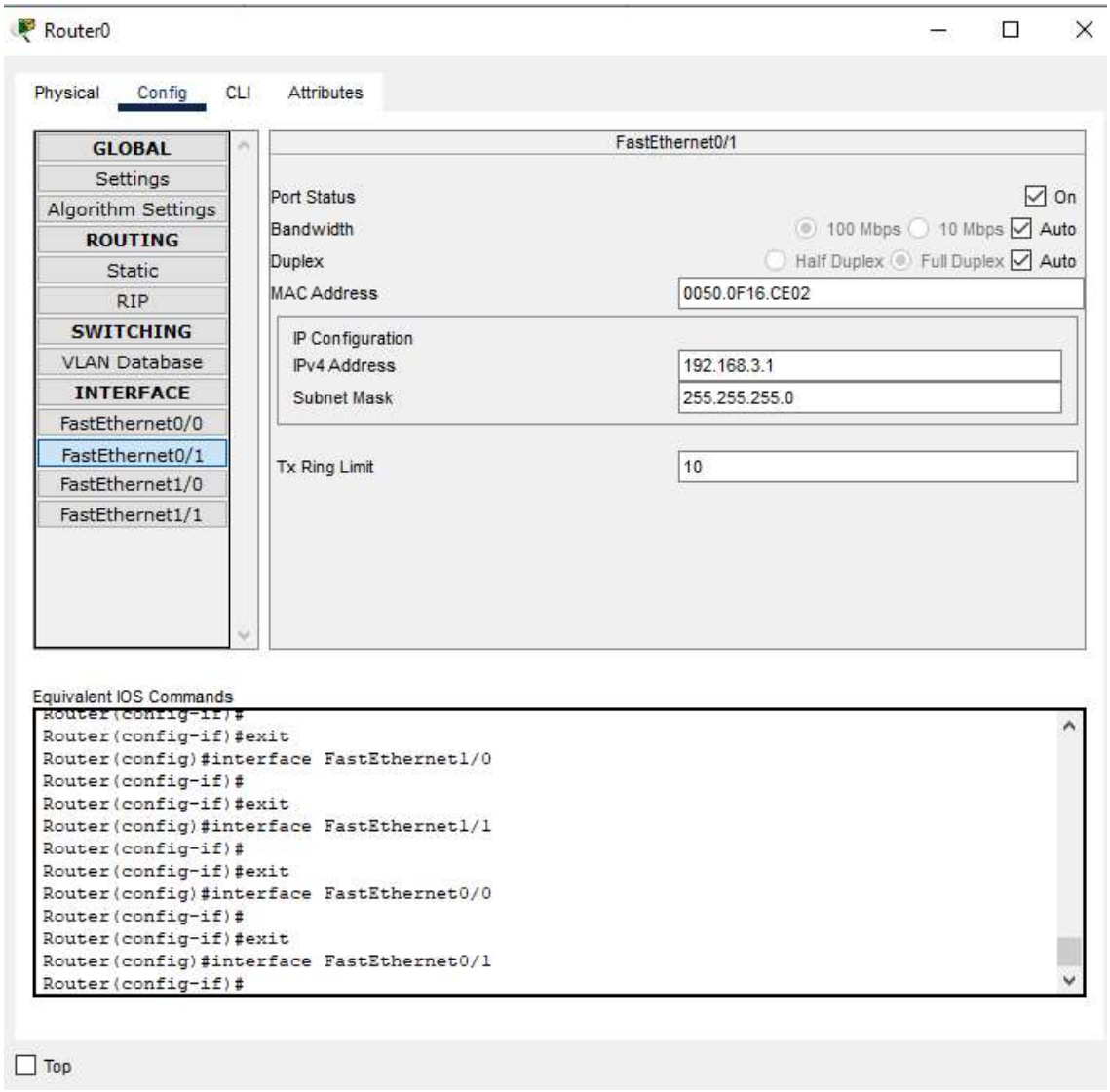
For example, For PC0:



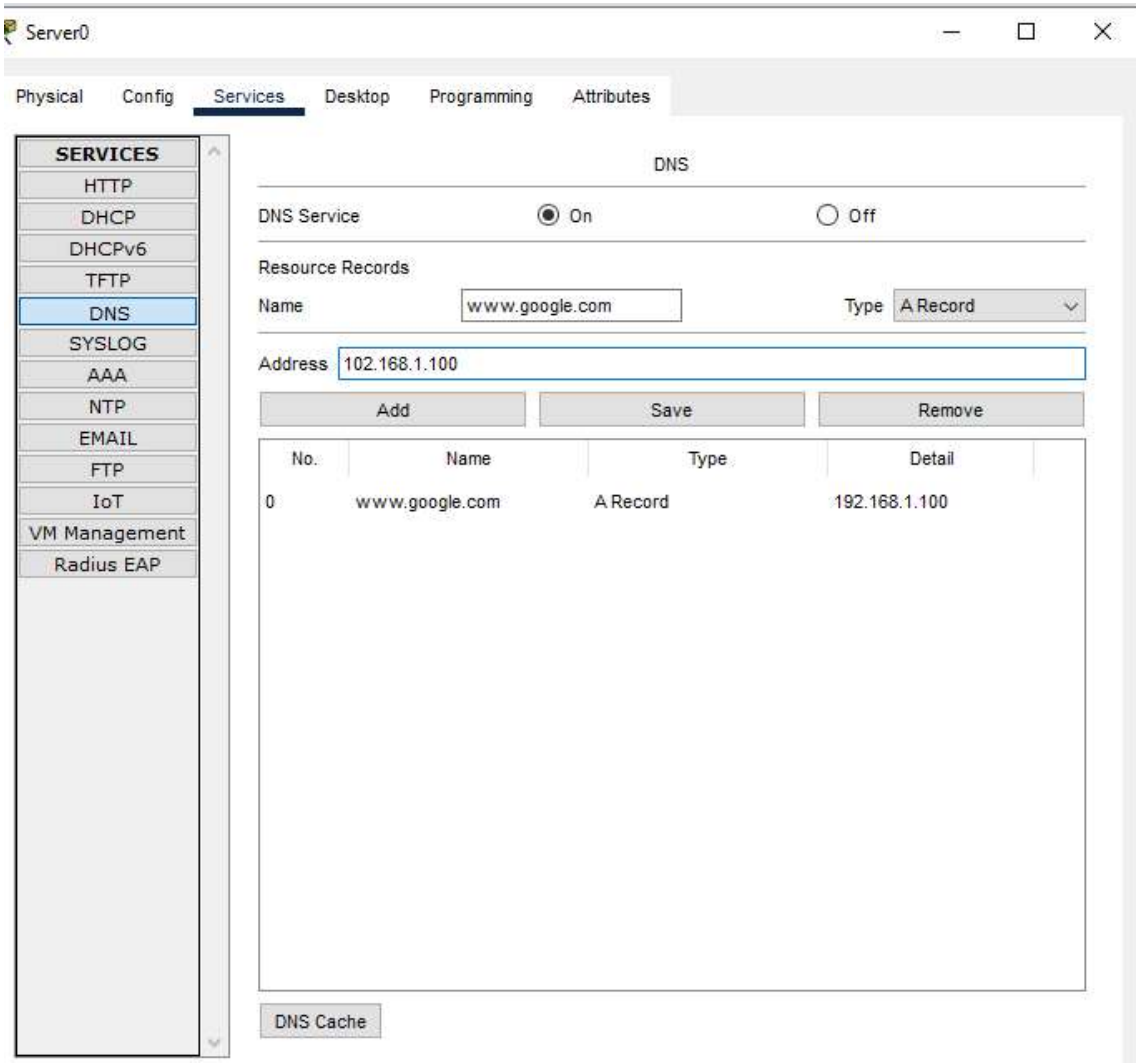
Step 5: Now, configure the router according to the details given below and then turn on the port status.

FastEthernet0/0	192.168.2.1	255.255.255.0
FastEthernet0/1	192.168.3.1	255.255.255.0
FastEthernet1/0	192.168.1.1	255.255.255.0
FastEthernet1/1	192.168.4.1	255.255.255.0

For example, For FastEthernet0/1:



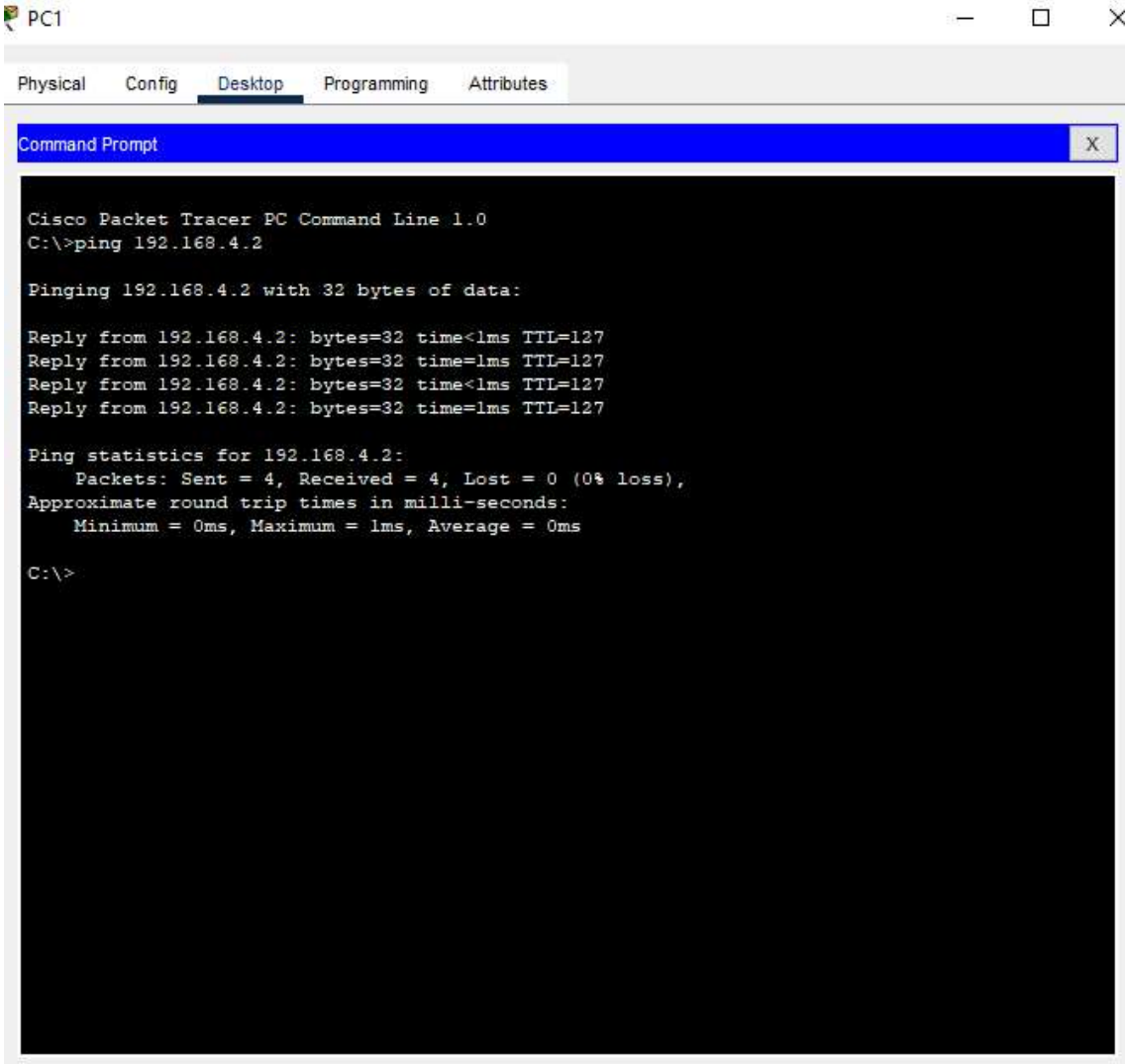
Step 6: Now, we've to maintain the DNS server. For this click on the server, go to the services section, and then click on DNS. Turn on the DNS server, Enter any domain in the 'Name' section For eg: "www.google.com" enter the IP address of the server in the 'Address section', and then click on save.



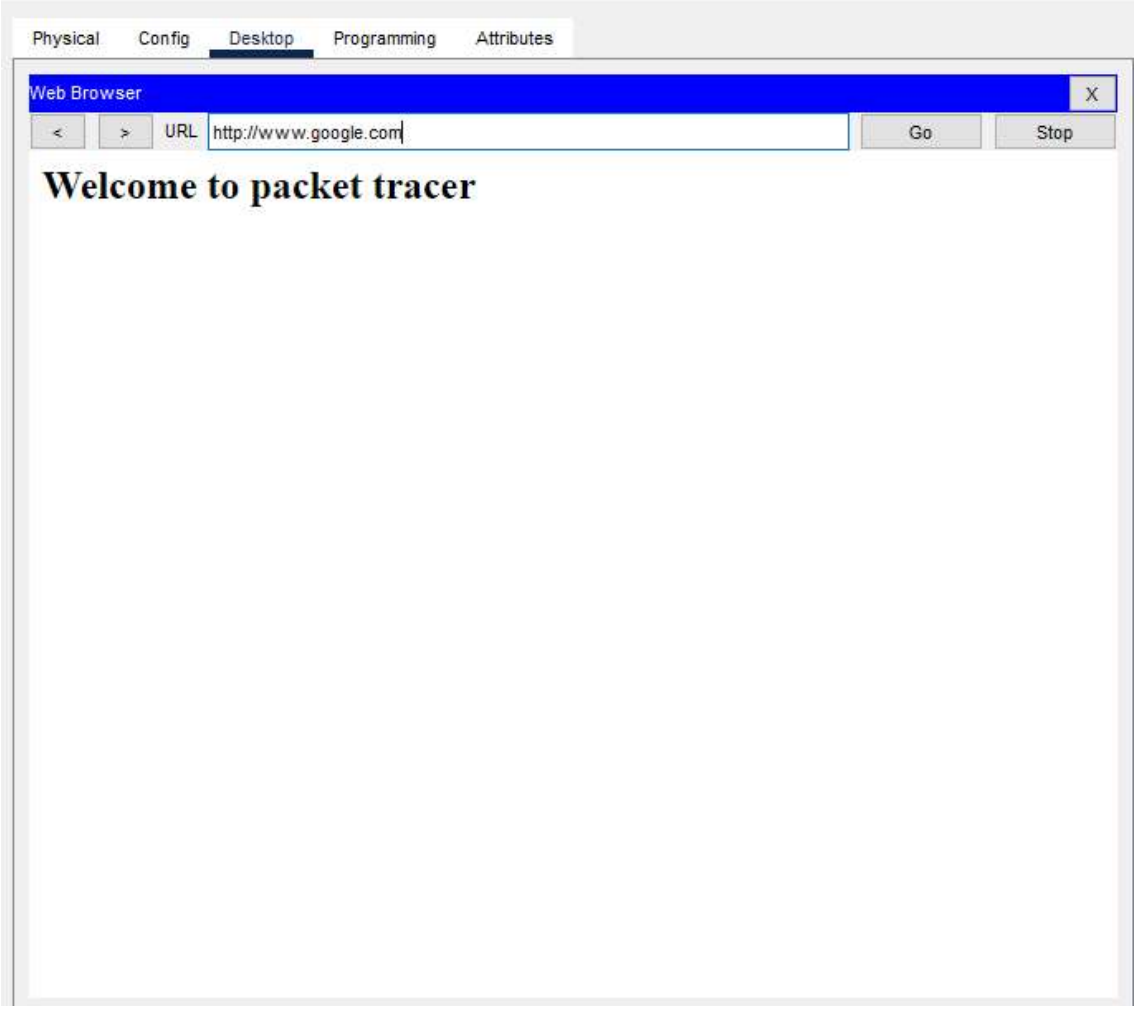
Step 7: Again click on the server, go to services, and then click on HTTP. Turn on the HTTP and HTTPS services. You can also edit the index.html file which will show when you search the domain you entered in DNS in the web browser.

Step 8: To check the connection between PCs present in different networks in this organization, you can use the ping command. Click on any PC, go to Desktop, and then click on the command prompt, enter the ping command, and check if they're able to communicate with each other.

For example, ping 192.168.3.1



Step 9: Click on any PC, go to the Desktop section, and then open the web browser. Enter the domain or address you've given in the DNS server and it will show the index.html file from HTTP services.


Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://www.google.com
Go
Stop
Welcome to packet tracer